Zoe Krauss

zkrauss@uw.edu
zoekrauss.github.io
(847)-873-4056

EDUCATION

Ph.D. University of Washington, School of Oceanography — 2024
Marine Geology & Geophysics; Advisor: William S. D. Wilcock

M.S. University of Washington, School of Oceanography — 2021

B.A. Colorado College — 2019
Physics major: Geophysics emphasis; Magna cum laude, Phi Beta Kappa

APPOINTMENTS

Postdoctoral Researcher, Pacific Northwest Seismic Network — 2025-present

Research and Teaching Assistant, University of Washington — 2019-2024

PUBLICATIONS

Krauss, Z., Mazur, M., Fontaine, N. K., Ryf, R., Neilson, D. T., Dienstfrey, W., et al. (accepted, in press at TSR). Distributed acoustic sensing beyond optical repeaters offshore Central Oregon: a community dataset for a transformative technology. Preprint: https://doi.org/10.22541/essoar.15004068/v1

Krauss, Z., Hartog, J. R., Dybing, S., Thompson, M., Wilcock, W. S.D., Dunham, A. (in review at BSSA). Existing offshore infrastructure could improve earthquake early warning in the U.S. Pacific Northwest.

Moriarty, S. N., Neufeld, M., **Krauss, Z**., & Jamieson, J. W. (2025). Geological, geophysical, and geobiological investigation of the inactive Nawaay'as hydrothermal vent field at West Valley, Juan de Fuca Ridge. Marine Geology, 107643.

Krauss, Z., Wilcock, W., & Creager, K. (2025). Possible Shallow Tectonic Tremor Signals Near the Deformation Front in Central Cascadia. Seismica, 2(4).

Krauss, Z., Baillard, C., Wilcock, W. S., Heesemann, M., Schlesinger, A., & Kukovica, J. (2025). A Single-Station Earthquake Catalog for the Endeavour Segment of the Juan de Fuca Ridge (2011– 2016): Challenges and Implications for the Spreading Cycle. Seismological Research Letters.

Evans, G. N., **Krauss, Z.,** Lilley, M. D., Seyfried Jr, W. E., & Wilcock, W. S. (2025). Tectonically induced changes in vent fluid compositions and metal concentrations at Main Endeavour Field, northeast Pacific Ocean. Earth and Planetary Science Letters, 666, 119485.

Denolle, M., **et al.** (2025). Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing through Diverse Training Modalities. Seismological Research Letters.

Krauss, Z., Ni, Y., Henderson, S., & Denolle, M. (2023). Seismology in the cloud: guidance for the individual researcher. Seismica, 2(2).

Krauss, Z., Wilcock, W. S.D., Heesemann, M., Schlesinger, A., Kukovica, J., & Farrugia, J. J. (2023). A Long-Term Earthquake Catalog for the Endeavour Segment: Constraints on the Extensional Cycle and Evidence for Hydrothermal Venting Supported by Propagating Rifts. Journal of Geophysical Research: Solid Earth, 128(2), e2022JB025662.

Krauss, Z., & Menke, W. (2020). The Northern Gulf Anomaly: P-and S-wave travel time delays illuminate a strong thermal feature beneath the Northern Gulf of Mexico. Earth and Planetary Science Letters, 534, 116102.

Almendros, J., **et al**. (2020). BRAVOSEIS: Geophysical investigation of rifting and volcanism in the Bransfield strait, Antarctica. Journal of South American Earth Sciences 104: 102834.

Borella, J., Quigley, M., **Krauss, Z.,** Lincoln, K., Attanayake, J., Stamp, L., Lanman, H., Levine, S., Hampton, S. & Gravley, D. (2019). Geologic and geomorphic controls on rockfall hazard: how well do past rockfalls predict future distributions?. Natural Hazards and Earth System Sciences, 19(10), 2249-2280.

ARTICLES IN PREPARATION

Krauss, Z., & Wilcock, W. S.D. (completed dissertation chapter). Repeating earthquakes and earthquake swarms at the Endeavour segment: direct evidence for aseismic slip at a mid-ocean ridge.

WHITE PAPERS

Krauss, Z., Ledeczi, A., Shuck, B., & Wei, M. (2026). Offshore Observations Special Interest Group Topical Workshop Report, October 27, 2025. CRESCENT Open Documents Library. https://doi.org/10.5281/ZENODO.20496895

Krauss, Z., Eilon, Z., Parnell-Turner, R., Janiszewski, H., Worthington, L., Kidiwela, M., & Brunsvik., B. (2021). Call to expand ocean bottom seismograph (OBS) facilities and instrument pool for ambitious Rift2Ridge science. 2021 Rift2Ridge Workshop.

INVITED TALKS

AGU Fall Meeting, San Francisco, CA, 2026.
Fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: the potential for long-term oceanographic, earthquake, tsunami, and acoustic monitoring.

InterRidge Webinar Series, 2024 (virtual seminar).
Near-real-time Seismic Monitoring of a Mid-ocean Ridge.

AGU Fall Meeting, San Francisco, CA, 2023 (talk).
Near-real-time Seismic Monitoring of Tectonically-Influenced Mid-ocean Ridge Hydrothermal Vent Fields.

USGS Menlo Park and Stanford Reading Group, 2021 (virtual seminar).
Deep structure of the Northern Gulf of Mexico from Seismic Data.

AWARDS

SSA Annual Meeting Student Presentation Award	2024
AGU Outstanding Student Presentation Award (OSPA)	2023
NDSEG Conference Exemplary Poster Award	2022
AGU Outstanding Student Presentation Award (OSPA)	2021
National Defense Science and Engineering Fellowship (NDSEG)	2020
Full tuition and $114,000 stipend support over 3 years	
Travel allowance of $5,000	
NSF Graduate Research Fellowship	2020
Declined in favor of the NDSEG	
ARCS Fellowship, University of Washington	2019

$17,500 of additional support over 3 years

NSF Graduate Research Fellowship, Honorable Mention 2019

Association of Women Geoscientists Outstanding Student Award 2019

Cowperthwaite Award for Excellence in Physics 2019
Awarded by the Colorado College Physics Department

RECENT CONFERENCE PRESENTATIONS

SSA Annual Meeting, Pasadena, CA (talk) 2026
DAS Beyond Repeaters: Multi-Span Monitoring using Submarine Cables in the Pacific Ocean

SSA Annual Meeting, Pasadena, CA (talk) 2026
Exploring the Use of Existing Offshore Cabled Infrastructure for Earthquake Early Warning in the Pacific Northwest

AGU Fall Meeting, New Orleans, LA (poster) 2025
Leveraging Existing Offshore Infrastructure for Earthquake Early Warning in the Pacific Northwest

SSA Annual Meeting, Baltimore, MD (poster) 2025
Potential tectonic tremor detected on a single ocean bottom seismometer offshore Cascadia

SSA Annual Meeting, Anchorage, AK (talk) 2024
Searching for Low-Amplitude Shallow Tectonic Tremor in Cascadia Using Buried Ocean Bottom Seismometers

SERVICE

Member of AGU Tectonophysics Early Career Committee 2026-

Session co-convenor, SSA Topical Meeting for Optical Seismology. 2026
Integration of Diverse Sensor Networks.

Guest editor, SRL Focus Section on Optical Seismology 2026

Session co-convenor, SSA 2026. *Advancing Earthquake Early Warning: Science, Technology, and Engagement in the U.S. and Beyond.* 2026

Member of the ANSS DAS working group 2026-

UW Seismolunch Seminar and Reading Club organizer 2025-

Session co-convenor, AGU 2025. The Cascadia Megathrust and Beyond: *From Fundamental Scientific Processes to Societal Resilience.* 2025

Co-organizer, SZ4D AGU Mini-Workshop 2025. *Open DAS Data for Subduction Zone Environments.* 2025

Organizing committee member, CRESCENT Offshore Observations Special Interest Group meeting. 2025

Organizing committee member, Endeavour 2024 Workshop. Cross-disciplinary planning meeting for scientific response to an anticipated volcanic event. 2024

Session co-convenor, SSA 2024. *Leveraging Cutting-Edge Cyberinfrastructure for Large Scale Data Analysis and Education.* 2024

PEER REVIEW

Seismica, Geophysical Journal International, Geophysical Research Letters, Seismological Research Letters; Guest Editor: Seismological Research Letters

TEACHING AND MENTORSHIP

UW Applied Physics Laboratory Summer Undergraduate Research Program (SURP) 2025
Primary mentor for undergraduate student Lydia Kelley
- Project continued through 2025-2026 as a senior thesis project
- Project produced a talk at AGU 2025 Annual Meeting

OCEAN 320: Coastal Oceanography 2024
Teaching assistant, University of Washington
- Co-led lab sessions, including some in Python
- Led office hours

Data Science in Oceanography Summer Program 2022
Mentor, University of Washington
- Prepared and led Python tutorials on introductory seismic analysis
- Gave introductory lecture to Marine Geology and Geophysics
- Served as mentor for an undergraduate project

OCEAN 201: Intro to Oceanography Lab 2021
Teaching assistant, University of Washington, virtual due to COVID
- Prepared lab experiments and reports
- Graded ~40 lab reports per week
- Co-led synchronous lab sessions

FIELD EXPERIENCE

OBS redeployment at Axial Seamount, R/V Sally Ride 2023
DAS deployment on OOI cabled network, Pacific City, OR 2021
OBS retrieval at Bransfield Strait, Antarctica, R/V Hesperides 2020
Frontiers Abroad geology field camp, New Zealand 2017

MEDIA COVERAGE

Graphics and interviews for Seattle Times coverage of "fanquakes"
- Seahawks 2026 playoffs run
- FIFA World Cup 2026

Live television interviews following the April 2026 Juan de Fuca Ridge earthquakes
- *KOMO news, FOX13 news*

News interviews following the March 2024 earthquake swarm at the Endeavour segment
- *Front page of the Times Colonist*
- *Additional stories by Vancouver Sun, CBC Canada, Live Science*

EarthScope interview for the Krauss et al. (2023) Seismica article
- *Cloud case study: a seismic processing tutorial for your first foray*